about the black hole, except the nature of the singularity at its center. For the singularity, they need the ill-understood laws of quantum gravity (Chapters 26).

A Black Hole's Appearance from Inside Our Universe

We humans are confined to our brane. We can't escape from it, into the bulk (unless an ultra-advanced civilization gives us a ride in a tesseract or some such vehicle, as they do for Cooper in *Interstellar*; see Chapter 29). Therefore, we can't see a black hole's warped space, as depicted in Figure 5.5. The black-hole funnels and whirlpools so often shown in movies, for example, Disney Studios' 1979 movie *The Black Hole*, would never be seen by any creature that lives in our universe.

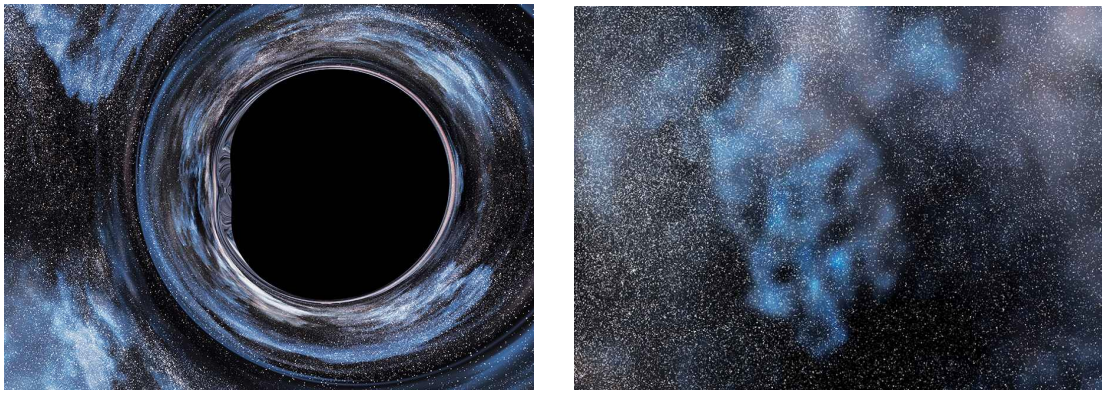


Fig. 5.6. A fast-spinning black hole (*left*) moving in front of the star field shown on the right. *[From a simulation for this book by the Double Negative visual-effects team.]*

Interstellar is the first Hollywood movie to depict a black hole correctly, in the manner that humans would actually see and experience it. Figure 5.6 is a example, not taken from the movie. The black hole casts a black shadow on the field of stars behind it. Light rays from the stars are bent by the hole's warped space; they are *gravitationally lensed*, producing a concentric pattern of distortion. Light rays coming